CHAPTER 1 · CORE THEORY

Introductory Remarks and Starting Assumptions

Fluid Mechanics — 2nd Year Bachelor's Degree in Aerospace Engineering (UC3M)

Reference sources: Notes.pdf (Chapter 1, pp. 1–8) | slides_Chapters1-2.pdf (1.0–1.8) · Profs. Antonio L. Sánchez & Javier Rodríguez-Rodríguez

1. Solids, Liquids, and Gases: Mechanical Behavior and Microscopic Structure

At the macroscopic level, the fundamental physical distinction between solids and fluids lies in their mechanical response to an applied tangential shear stress τ:

- **Solids:** Under a small external shear force, solids respond with a finite deformation. The internal restoring force is proportional to the **deformation**:

  $$\tau \propto d\theta$$

  The solid retains an intrinsic, fixed geometric shape.
- **Fluids (Liquids and Gases):** Deform continuously under the action of tangential shear stresses, no matter how small; they are fundamentally incapable of remaining at rest under shear. The internal resistive force is proportional to the **rate of strain**:

  $$\tau \propto \frac{d\theta}{dt}$$

  Consequently, fluids lack a preferred shape and conform to the geometry of their confining container.

Liquids vs. Gases: Density and Compressibility

The mechanical differences between liquids and gases are less fundamental and center on two properties:

- **Density:** $\rho_{\text{liquid}} \gg \rho_{\text{gas}}$. Due to their substantial mass per unit volume, the force required to accelerate a given volume of liquid is orders of magnitude larger than that needed for a gas.
- **Compressibility:** The increase in density caused by an isothermal overpressure is negligible in liquids compared to gases:

$$\left(\frac{\partial \rho}{\partial p}\right)_{T,\text{liquid}} \ll \left(\frac{\partial \rho}{\partial p}\right)_{T,\text{gas}} \qquad \text{[Notes.pdf, Eq. 1.1]}$$

Halving the specific volume of liquid water requires a colossal overpressure on the order of 10^6 atmospheres.

For this reason, under most engineering conditions, liquids can be modeled as fluids of constant density (**"perfect liquids"**).

Microscopic Origin: Intermolecular Forces and Mean Spacing (d)

These macroscopic properties are direct consequences of the microscopic structure of matter. The net intermolecular force $F(d)$ exhibits strong repulsion at very short ranges and attraction at larger distances, reaching stable equilibrium at a characteristic critical spacing $d_0 \sim 3 \times 10^{-10}\ \text{m} = 3\ \text{Å}$.

Given the density ρ, molar mass W, and Avogadro's number $N_A = 6.022 \times 10^{23}$ molecules/mol, the mass of a single molecule is W/N_A. Assuming each molecule occupies a cube of mean side d, we find:

$$\rho = \frac{W/N_A}{d^3} \implies \mathbf{d = \left(\frac{W}{\rho N_A}\right)^{1/3}}$$

Quantity	Air (Standard Sea-Level Gas)	Water (Liquid)
Density (ρ)	$1.2\ \text{kg/m}^3$	$1000\ \text{kg/m}^3$
Molar Mass (W)	$28.96\ \text{g/mol}$	$18.02\ \text{g/mol}$
Intermolecular Spacing (d)	$\mathbf{d \approx 3.4 \times 10^{-9}\ \text{m}} \approx 10\, d_0$	$\mathbf{d \approx 3.1 \times 10^{-10}\ \text{m}} \approx d_0$
Molecules in $1\ \text{mm}^3$	$\sim 10^{16}$ molecules	$\sim 10^{19}$ molecules
Dynamic Behavior	Free molecular flight; interactions occur solely via binary collisions.	Dense packing; permanent cohesive intermolecular forces.

2. The Continuum Hypothesis and the Concept of Fluid Particle

The vast number of molecules involved (10^{16} in 1 mm^3 of air) renders solving Newton's equations of motion for individual molecules utterly intractable. To describe the macroscopic flow, we adopt the **continuum hypothesis**.

The Macroscopic Scale (L)

We define the characteristic macroscopic length scale L as the spatial distance over which significant variations in flow properties occur (for example, in a room $L \sim 10$ cm; over an airfoil, $L \sim$ chord length).

The Fluid Particle (δV) and the Double-Bounding Condition

We subdivide the fluid into small volume elements δV centered at position $\vec{x}$ at time t. For macroscopic field variables to be uniquely defined, the size $(\delta V)^{1/3}$ must strictly satisfy:

$$\mathbf{d \ll (\delta V)^{1/3} \ll L} \qquad [\text{Notes.pdf, Eq. 1.2}]$$

- **Lower bound ($(\delta V)^{1/3} \gg d$):** The volume must contain millions of molecules so that statistical averaging is free from discontinuous fluctuations. Plotting apparent density $\sum m_i / \delta V$ versus volume size $(\delta V)^{1/3}$ shows abrupt statistical jumps for tiny volumes each time a molecule enters or leaves. Beyond the microscopic scale d, the density reaches a constant, volume-independent value: the **continuum plateau**.
- **Upper bound ($(\delta V)^{1/3} \ll L$):** The particle must be sufficiently small compared to the global flow dimensions so that it can be treated as a mathematical differential point at $\vec{x}$.

Validity Criterion of the Continuum Medium

The continuum hypothesis is rigorously applicable whenever the following condition holds:

$$\frac{d}{L} \ll 1$$

For air at standard atmospheric conditions, $d \approx 3.4$ nm, meaning virtually all aeronautical engineering flows satisfy this condition with wide margins. The continuum approximation **ceases to be valid** in highly rarefied flows (satellite atmospheric re-entry in the thermosphere at altitudes $h > 100$ km, where $d \sim L$) and in microelectromechanical systems (MEMS/nanofluidics).

3. Definition of Macroscopic Fields: Density, Velocity, and Internal Energy

From the concept of a fluid particle δV, the continuous macroscopic fields are formulated mathematically:

1. Density Field (ρ):

$$\rho(\vec{x}, t) = \lim_{\delta V \to 0,\, (\delta V)^{1/3} \gg d} \frac{\sum m_i}{\delta V} \qquad [\text{Eq. } 1.3]$$

2. Flow Velocity Field ($\vec{v}$):

Represents the center-of-mass velocity of the set of molecules contained within the fluid particle:

$$\vec{v}(\vec{x}, t) = \lim_{\delta V \to 0} \frac{\sum m_i \vec{v}_i}{\sum m_i} \qquad [\text{Eq. } 1.4]$$

3. Total Energy and Decomposition of Internal Energy (e):

The total specific energy per unit mass is decomposed exactly into the macroscopic kinetic energy of the flow and the thermal internal energy:

$$\lim_{\delta V \to 0} \frac{\sum E_i}{\sum m_i} = e(\vec{x}, t) + \frac{|\vec{v}|^2}{2} \qquad [\text{Eq. } 1.5]$$

where the **internal energy** e collects the kinetic energy of random molecular motion relative to the center of mass together with intramolecular vibrational ($E_{v,i}$) and rotational ($E_{r,i}$) energies:

$$\mathbf{e} = \lim_{\delta \mathbf{V} \to \mathbf{0}} \frac{\sum \mathbf{m_i} \frac{|\vec{\mathbf{v}}_\mathbf{i} - \vec{\mathbf{v}}|^\mathbf{2}}{\mathbf{2}} + \mathbf{E_{v,i}} + \mathbf{E_{r,i}} + \cdots}{\sum \mathbf{m_i}} \qquad [\text{Eq. } 1.6]$$

The term $\frac{|\vec{v}_i - \vec{v}|^2}{2}$ corresponds to disordered thermal agitation, a quantity directly tied to the microscopic definition of **temperature** (T).

4. Local Thermodynamic Equilibrium Hypothesis (LTE)

Classical equilibrium thermodynamics assumes homogeneous systems with invariant properties. In fluid mechanics, however, properties vary continuously across space and time. Nevertheless, equilibrium thermodynamic relations hold locally: an observer moving with the local fluid velocity $\vec{v}(\vec{x}, t)$ can describe the local thermal state using the very same **equations of state** that govern strict thermodynamic equilibrium.

The Physical Mechanism: Molecular Collisions

In gases, molecules continually exchange momentum and energy through collisions, rapidly adjusting their velocity distribution to the local state. This process determines two key microscopic scales:

- **Mean free path (λ):** The average distance traveled by a molecule between two successive collisions. Equating the volume swept out along its path ($d_0^2\lambda$) to the available volume per molecule in the gas (d^3):

$$\frac{\lambda}{d} \simeq \left(\frac{d}{d_0}\right)^2 \implies \lambda \approx \mathbf{4} \times \mathbf{10^{-7}}\ \mathrm{m} = 0.4\ \mu\mathrm{m} \quad \text{(sea-level air)}$$

- **Characteristic collision time (τ):** With $a \approx \sqrt{\gamma RT} \approx 340\ \mathrm{m/s}$ as the mean thermal speed (on the order of the speed of sound):

$$\tau = \frac{\lambda}{a} \approx \frac{4 \times 10^{-7}\ \mathrm{m}}{340\ \mathrm{m/s}} \approx \mathbf{10^{-9}}\ \mathrm{s} = 1\ \text{nanosecond}$$

Knudsen Number Criterion (Kn)

For a molecule to undergo thousands of collisions and achieve equilibrium before migrating into regions with noticeably different macroscopic properties, the mean free path must be negligible compared to the macroscopic scale L:

$$\mathbf{Kn} = \frac{\lambda}{\mathbf{L}} \ll \mathbf{1} \qquad [\text{Notes.pdf, Eq. 1.7}]$$

For unsteady flows with a characteristic macroscopic time scale T_{macro}, we analogously require $T_{\text{macro}} \gg \tau \approx 10^{-9}$ s.

> **Fundamental Remark on the Knudsen Number**
>
> The Local Thermodynamic Equilibrium criterion ($Kn = \lambda/L \ll 1$) is **strictly more restrictive** than the continuum medium hypothesis ($d/L \ll 1$). In gases, the molecular mean free path is nearly two orders of magnitude larger than the intermolecular spacing ($\lambda \approx 400\ \mathrm{nm} \gg d \approx 3.4\ \mathrm{nm}$).

5. Thermodynamic Variables and Equations of State

Under the validity of LTE, the fundamental Gibbs thermodynamic relation links specific internal energy e, entropy s, and density ρ:

$$de = Tds - p\,d\left(\frac{1}{\rho}\right) = Tds + \frac{p}{\rho^2}d\rho \qquad \text{[Eq. 1.8]}$$

From this relation, the operational definitions of temperature, pressure, and specific enthalpy h follow:

$$T = \left(\frac{\partial e}{\partial s}\right)_{\rho} \quad \text{[Eq. 1.9]}, \qquad p = -\left(\frac{\partial e}{\partial \rho^{-1}}\right)_{s} \quad \text{[Eq. 1.10]}, \qquad h = e + \frac{p}{\rho}$$

The course syllabus formulates two fundamental canonical models for aerospace engineering:

A. Perfect Liquid Model (Incompressible)

A perfect liquid is characterized by **constant density** ρ_0 and **constant specific heat** c:

$$\rho = \rho_0 \qquad \text{[Eq. 1.11]}$$

$$\mathbf{e} = \mathbf{cT} + \mathbf{e_0} \qquad \text{[Eq. 1.12]}$$

$$\mathbf{h} = \mathbf{cT} + \mathbf{e_0} + \frac{\mathbf{p}}{\rho_0} \qquad \text{[Eq. 1.13]}$$

$$\mathbf{s} = \mathbf{c}\ln(\mathbf{T}) + \mathbf{s_0} \qquad \text{[Eq. 1.14]}$$

where e_0 and s_0 are reference constants. In the vast majority of hydraulic and liquid propellant systems, water is modeled as a perfect liquid with $\rho_0 = 1000\ \mathrm{kg/m^3}$ and $c = 4180\ \mathrm{J/(kg \cdot K)}$.

B. Perfect Gas Model (Calorically Perfect)

The thermal equation of state for ideal gases relates pressure, density, and temperature:

$$\frac{\mathbf{p}}{\rho} = \mathbf{R_g T} \qquad [\text{Eq. } 1.15]$$

$R_g = R_0 / W$
where the specific gas constant is , with $R_0 = 8.314\ \text{J}/(\text{mol}\cdot\text{K})$ being the universal gas constant and W the mean molecular mass.
The caloric equations of state for internal energy, enthalpy, and entropy are:

$$\mathbf{e} = \mathbf{c_v T} + \mathbf{e_0} \qquad [\text{Eq. } 1.16]$$

$$\mathbf{h} = \mathbf{c_p T} + \mathbf{e_0} \qquad [\text{Eq. } 1.17]$$

$$\mathbf{s} = \mathbf{c_v} \ln\left(\frac{\mathbf{p}}{\rho^{\gamma}}\right) + \mathbf{s_0} \qquad [\text{Eq. } 1.18]$$

where c_v and c_p are the specific heats at constant volume and constant pressure, linked by **Mayer's relation**:

$$c_p = c_v + R_g$$

The heat capacity ratio $\gamma = c_p / c_v$ is $\gamma = \mathbf{7/5} = \mathbf{1.4}$ for diatomic gases (N_2, O_2, air). For standard dry air in subsonic and moderate supersonic regimes:

$$R_g = 287\ \text{J}/(\text{kg}\cdot\text{K}), \quad c_v = 717\ \text{J}/(\text{kg}\cdot\text{K}), \quad c_p = 1004\ \text{J}/(\text{kg}\cdot\text{K})$$

Model limits: At extreme pressures or cryogenic temperatures, real gas equations of state (e.g., Van der Waals) must be used. Furthermore, when temperature variations are substantial (e.g., combustion chambers or rocket nozzles), specific heats cease to be constant and become functions of temperature $c_p(T), c_v(T)$.

6. Summary of Formulas and Orders of Magnitude

Concept / Quantity	Mathematical Expression	Typical Values (Sea-Level Air)
Intermolecular Spacing	$d = \left(\frac{W}{\rho N_A}\right)^{1/3}$	$d \approx 3.4 \times 10^{-9}\ \text{m} = 3.4\ \text{nm}$
Fluid Particle Condition	$d \ll (\delta V)^{1/3} \ll L$	$3.4\ \text{nm} \ll (\delta V)^{1/3} \ll 10\ \text{cm}$
Continuum Hypothesis Criterion	$\frac{d}{L} \ll 1$	$\frac{d}{L} \sim 10^{-8} \lll 1$ (broad margin)
Mean Free Path	$\frac{\lambda}{d} \simeq \left(\frac{d}{d_0}\right)^2$	$\lambda \approx 4 \times 10^{-7}\ \text{m} = 0.4\ \mu\text{m}$
Mean Collision Time	$\tau = \frac{\lambda}{a}$	$\tau \approx 10^{-9}\ \text{s} = 1\ \text{ns}$
Knudsen Criterion (LTE)	$Kn = \frac{\lambda}{L} \ll 1$	$Kn \sim 10^{-6} \ll 1$ (more restrictive than d/L)
Perfect Gas Equation	$p = \rho R_g T$	$R_g = 287\ \text{J}/(\text{kg}\cdot\text{K}), \gamma = 1.4$
Perfect Liquid Specific Heat	$e = cT + e_0, \quad \rho = \rho_0$	Water: $\rho_0 = 1000\ \text{kg}/\text{m}^3, c = 4180\ \text{J}/(\text{kg}\cdot\text{K})$